

cryptography-journal / week03.md

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Week 0 3 - Cryptography Journal

Algorithm: DES in ECB Mode

Objective: Encrypt plaintext using the Data Encryption Standard (DES) in Electronic Codebook (ECB) mode using OpenSSL.

Commands and Execution

Check OpenSSL version

openssl version

Create plaintext file

echo "This is a 24-byte sample text." > plaintext.txt

Encrypt using DES in ECB mode

openssl enc -des-ecb -e -in plaintext.txt -out ciphertext_ecb.bin -K 0123456789abcdef -provider legacy

View encrypted output using hexdump

xxd ciphertext_ecb.bin

Output:

e824 8aa4 db58 5b12 5eed 1a1d dc60 1ab4

fa89 205c 4896 cc31 d25e 77bc 72ad 47f6

Error Handling with No Padding

Attempted No Padding Encryption:

openssl enc -des-ecb -e -in plaintext.txt -out ciphertext_no_padding.bin -K 0123456789abcdef -nopad -provider legacy Error:

bad encrypt

error:1C800068:Provider routines:ossl_cipher_generic_block_finalwrong final block length

- Reason: DES requires input in 8-byte blocks. Without padding, encryption fails if the input is not a multiple of 8 bytes.

Modified Plaintext Attempt

echo "Thiz is a 24-byte sample text." > plaintext_modified.txt

openssl enc -des-ecb -e -in plaintext_modified.txt -out ciphertext_modified.bin -K 0123456789abcdef -nopad -provider legacy

- Still resulted in a block length error.

Algorithm: AES in CBC Mode

Objective: Encrypt plaintext using AES-128 in Cipher Block Chaining (CBC) mode.

Commands and Execution

Encrypt using AES-128-CBC

openssl enc -aes-128-cbc -e -in plaintext.txt -out ciphertext_aes.bin \

-K 00112233445566778899aabbccddeeff -iv 0102030405060708

Output:

c706 0fcf 8bb2 561b 757a e8d1 12a0 7c8d

3030 bccd 0c3d a9e1 4178 553d 4194 ff8f

Performance Evaluation

openssl speed aes-128-cbc

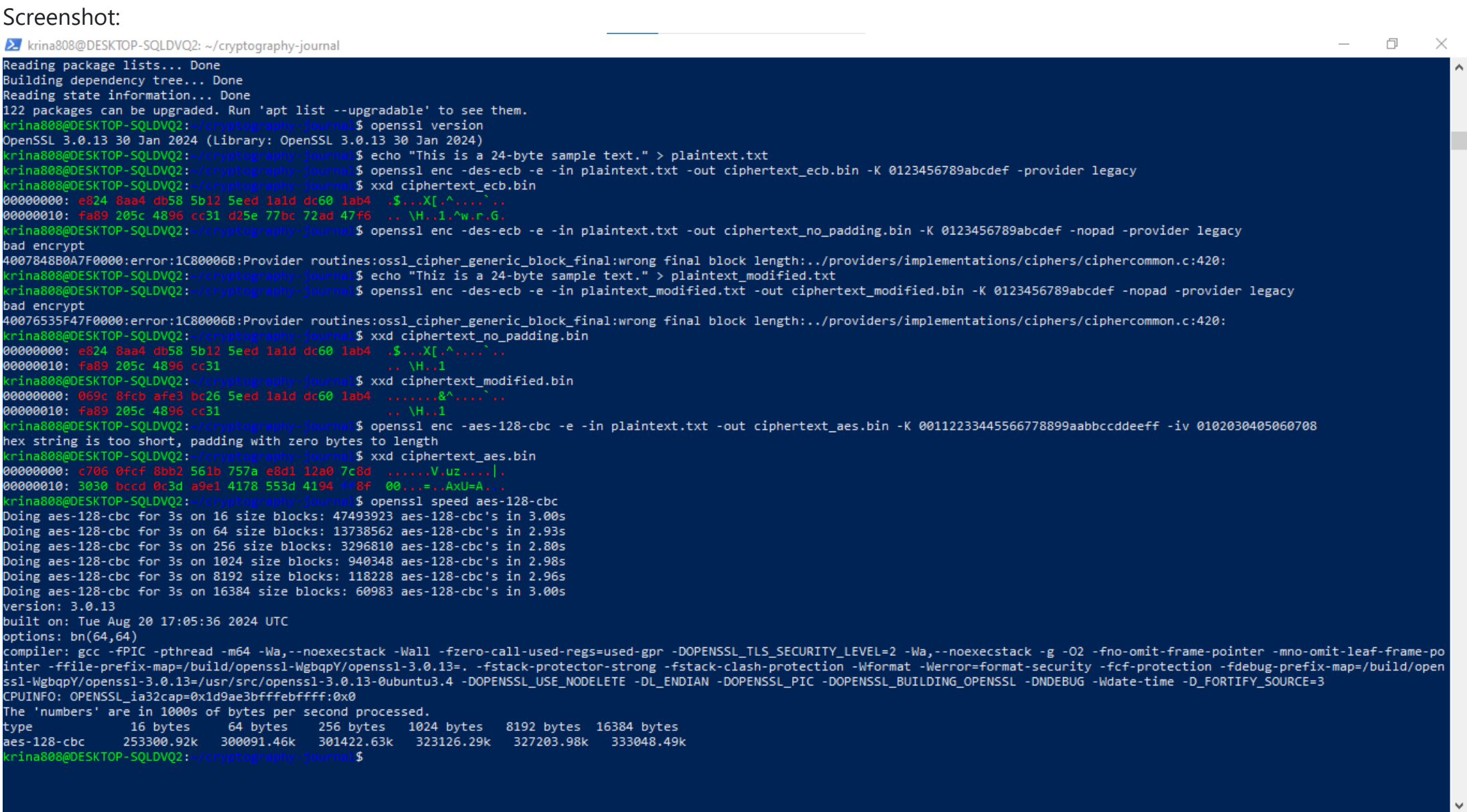
Output:

Doing aes-128-cbc for 3s on 16 size blocks: 47493923 aes-128-cbc's in 3.00s

Doing aes-128-cbc for 3s on 64 size blocks: 13738562 aes-128-cbc's in 2.93s

Doing aes-128-cbc for 3s on 256 size blocks: 3296810 aes-128-cbc's in 2.80s

Doing aes-128-cbc for 3s on 1024 size blocks: 940348 aes-128-cbc's in 2.98s



Attacks on Systems:

DES using ECB mode is shown to be vulnerable to pattern attacks, because of identical plaintext blocks resulting in identity ciphertexts which allow you to deduce information. For those attack, AES in CBC mode has used Initialization Vector (IV) which makes it more resistant to such attacks.

Difficulties in Understanding Security Systems:

Initially DES encryption did not make too much sense for me because neither did I know that DES encryption takes 8 byte blocks. I only understood it after getting the concept of padding and how the IV plays a role in AES CBC mode.

Links to Security Systems:

OWASP Link: <https://owasp.org/>

Summary: Web applications security is a complicated issue and while there are excellent guides, book and articles out there for this area, there is no single source to provide guidance on how to secure a web app. It consists of guides, tools and frameworks to protect organizations' systems from common vulnerabilities like SQL injection, cross site scripting (XSS) and broken authentication among other things. The OWASP Top 10, a well-known list of the most important web application security issues, is a must for developers and security professional.

Insights and Reflections

AES in CBC mode is more secure and efficient compared to DES in ECB mode, as it introduces randomness using an Initialization Vector (IV) and prevents identical ciphertext for repeated plaintext. The padding issue observed in DES encryption highlights the importance of ensuring plaintext sizes are aligned with block sizes. Additionally, AES-128 demonstrated superior performance, making it a preferred choice for modern encryption needs.

